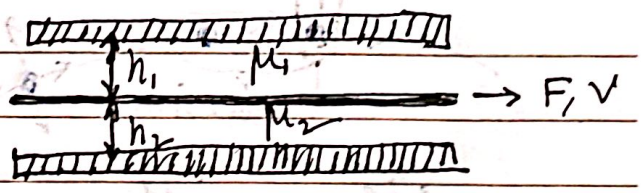
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Assignment 2.1

A thin plate is separated from two fixed plates by very viscous liquids μ_1 and μ_2, respectively, as in Fig 2.1. The plate spacings h_1 and h_2 are unequal, as shown. The contact area A is between the center plate and each fluid

(a) Assuming linear velocity distribution in each fluid derive the force F required to pull the plate at velocity V

(b) Is there any relation between two viscorities ?



Sol^n

(a) $u_1 = ay + b$

$u_2 = cy + d$

$\tau_1 = \mu_1 \frac{\partial}{\partial y}(ay + b)$

$= \mu_1 a$

Similarly $\tau_2 = \mu_2 c$

$0 = a \times 0 + b$

$b = 0$

$v = a \times h_1 + 0$

$a = v/h_1$

$c = v/h_2$

$F = A(\tau_1 + \tau_2) = \mu_1 a + \mu_2 c$

$= \left(\frac{\mu_1}{h_1} + \frac{\mu_2}{h_2}\right) v A$

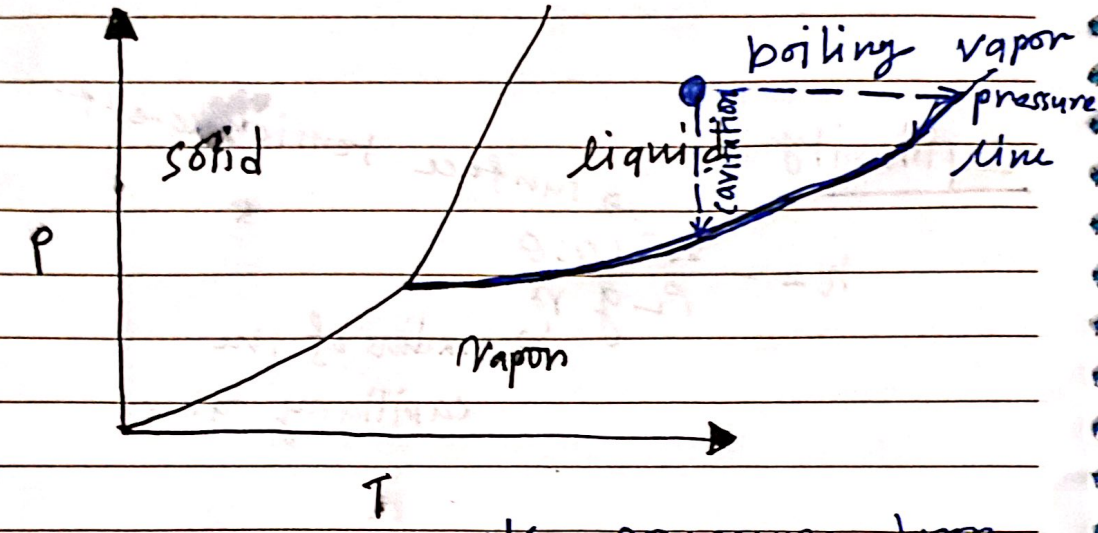
(b) No relation ☹

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Unit: N/m^2 or Pa

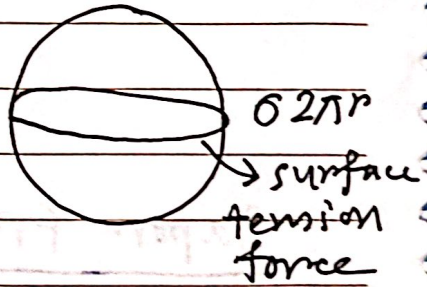
Phase diagram



* pressure drop causes cavitation
↳ vapor formation

Assignment 2.2

Young-Laplace Eqn For a sphere



$$F_{in} - F_{out} = F_{surface}$$

$$P_{in}\pi r^2 - P_{out}\pi r^2 = 2\pi r\sigma$$

$$\Rightarrow P_{in} - P_{out} = \frac{2\pi r\sigma}{\pi r^2} = \frac{2\sigma}{r}$$

HEART'S